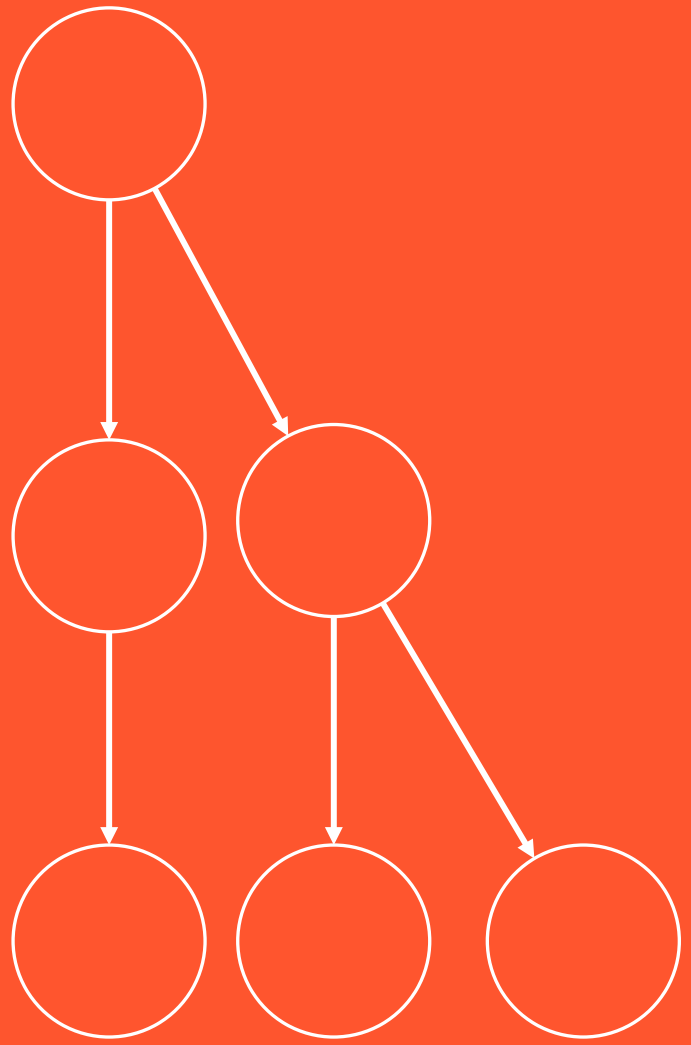


BCOG 100

Planning and Decision-making

Lecture 12.2 — Searching, Choosing, and General Ability

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Did the Jays Plan?

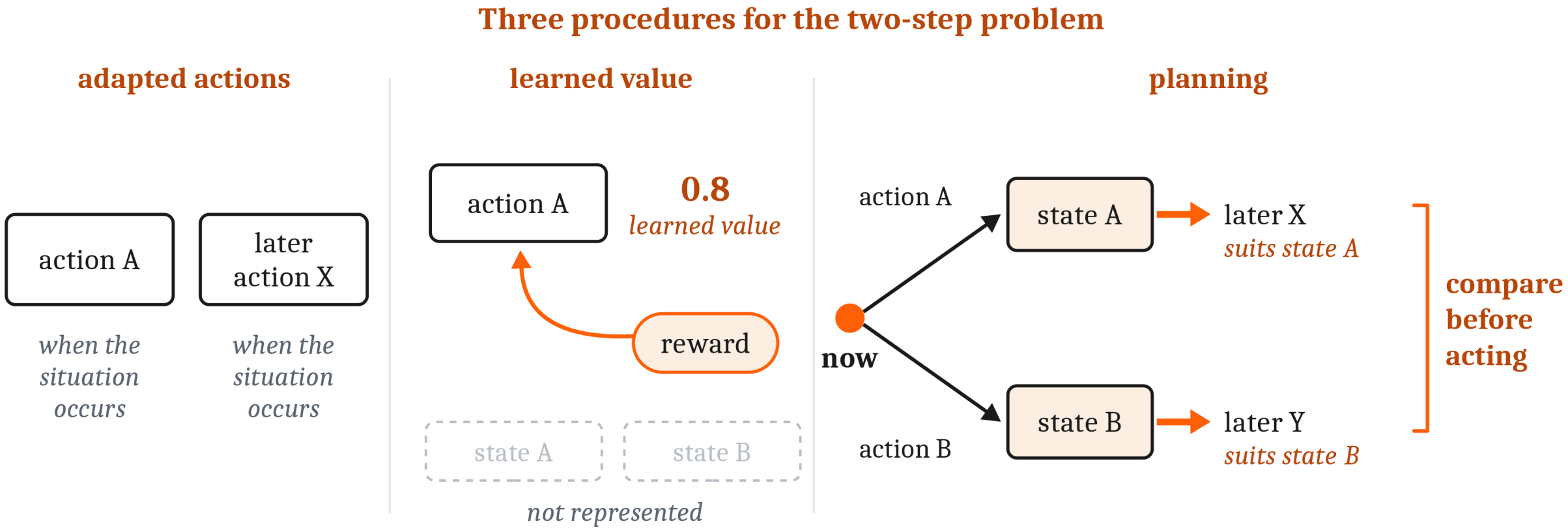
In your group, three minutes.

Scrub jays cached more nuts in the compartment where they had gone without breakfast.

What would it take to show that they planned, rather than repeating what had been rewarded before?

Three Procedures, One Behavior

Model-free: choosing an action by repeating what has worked in that situation before, with no model of what the action leads to. **Model-based**: choosing an action by running a model of what leads to what, and selecting on the basis of where the model ends up. **Planning**: choosing a sequence of actions by working through their predicted consequences before acting.



Telling the Procedures Apart

All three procedures can produce the same behavior in a familiar situation. Telling them apart takes a test that the first two would fail.

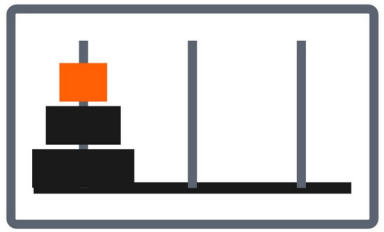
Procedure	What the animal represents	When the coming situation differs from what usually follows
Adapted actions	Nothing about the later situation; each action is attached to its own situation	Its actions suit only the situation that usually follows
Learned value (model-free)	The value of the first action, not the state it brings about	The choice cannot change with the coming situation
Planning (model-based)	The state each first action brings about, and the later action that suits each state	Only planning meets this harder version of the problem

Problems as Spaces to Search

A **problem space** is the set of states a problem can be in, as the solver represents them, with a starting state, a goal, and moves that lead from one state to another. **Search**: trying sequences of moves in a problem space to find one that reaches the goal. A **heuristic** is a rule of thumb that directs search toward small, promising parts of a problem space, so that the rest can be ignored.

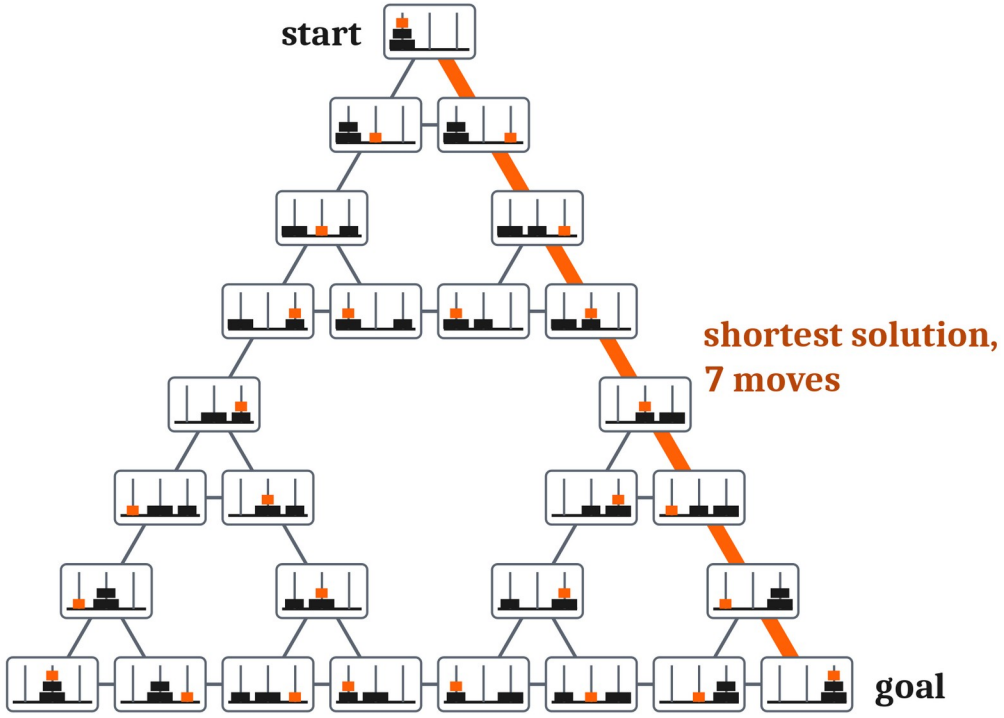
Every arrangement of the three-disk Tower of Hanoi

each box is an arrangement;



left middle right

this one is the start:
all three disks on the left peg



computed for this chapter: 27 arrangements, 39 legal moves

Search in Machines and on Maps

Game-playing programs choose moves by searching ahead, guided by estimates they have learned.

AlphaZero's values are learned from experience, as in model-free choice. Its search runs through positions predicted from the rules, as in model-based choice.

MuZero searched through a representation of the game that it had learned. Choosing a route on a map is search in a problem space whose states are places.

Limiting Search

A large loss roughly doubled the chance that people stopped searching down that branch, even when a larger gain lay beyond it.

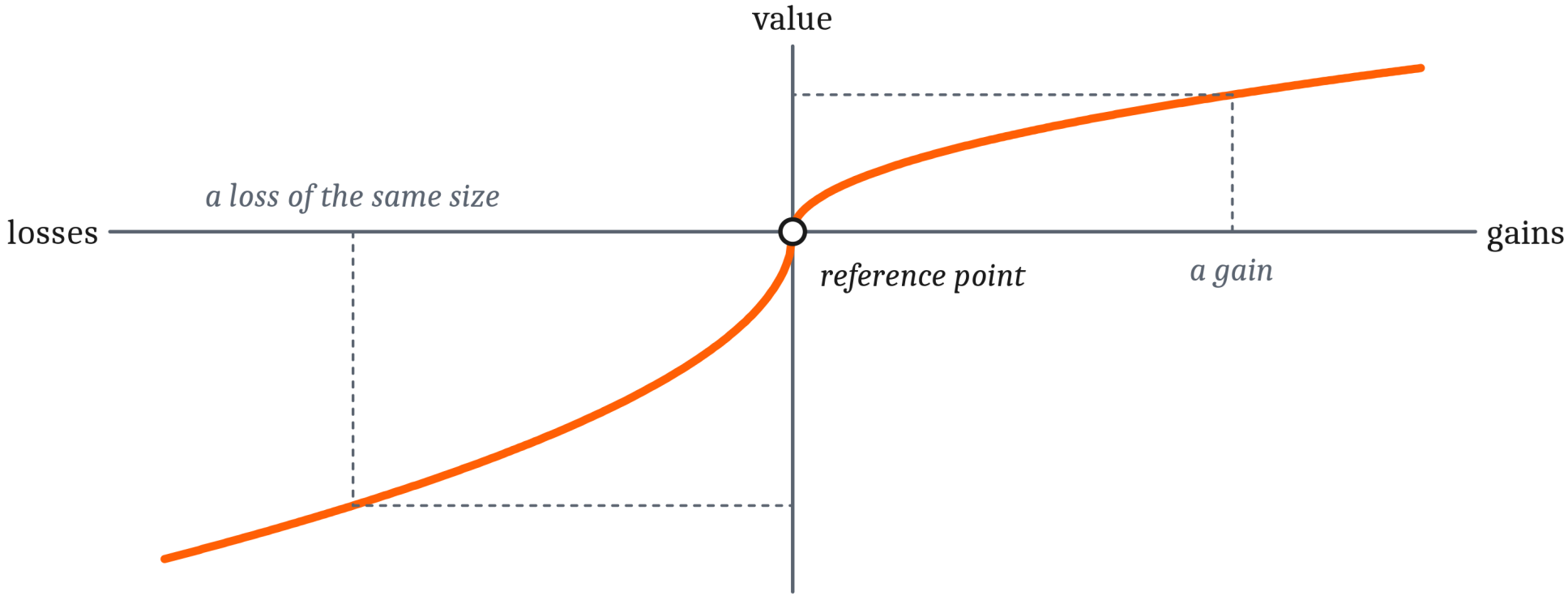
On a *resource-rational* account, a searcher weighs what further thinking would gain against the time or points that each step of thinking uses up.

How an agent chooses between planning and habit remains open.

Choosing Under Risk

Bounded rationality: the view that choices are made by procedures suited to a chooser's limited knowledge, time, and ability to compute. **Reflection effect**: the reversal of attitudes toward risk when the same choices are put as losses rather than gains. **Loss aversion**: the tendency for a loss to count for more than a gain of the same size. **Framing effect**: a change in choice when the same options are described differently.

A value function, as proposed in prospect theory



after Kahneman and Tversky (1979), Figure 3, p. 279; hypothetical, so drawn without numbers

Waiting

Delay discounting: the tendency for a reward to count for less the longer one must wait for it. **Self-control**: acting for a later or larger outcome against a present temptation.

Waiting as	What the evidence shows
A choice	A reward counts for less the longer the wait, so waiting means choosing the larger reward despite the delay
Partly something a child does	Children waited longer with the treats out of sight; the longest waiters sang, talked to themselves, or invented games
Partly a belief about the world	Children waited about 3 minutes after an experimenter broke her promises, and about 12 after she kept them

Executive Function

Executive function: the set of processes that control other mental processes in the service of a goal, needed when a habit or the most obvious response would not work. Three components are commonly named: *inhibition*, *shifting*, and *updating* what is in **working memory**, which is memory for what a task needs while the task is being done.

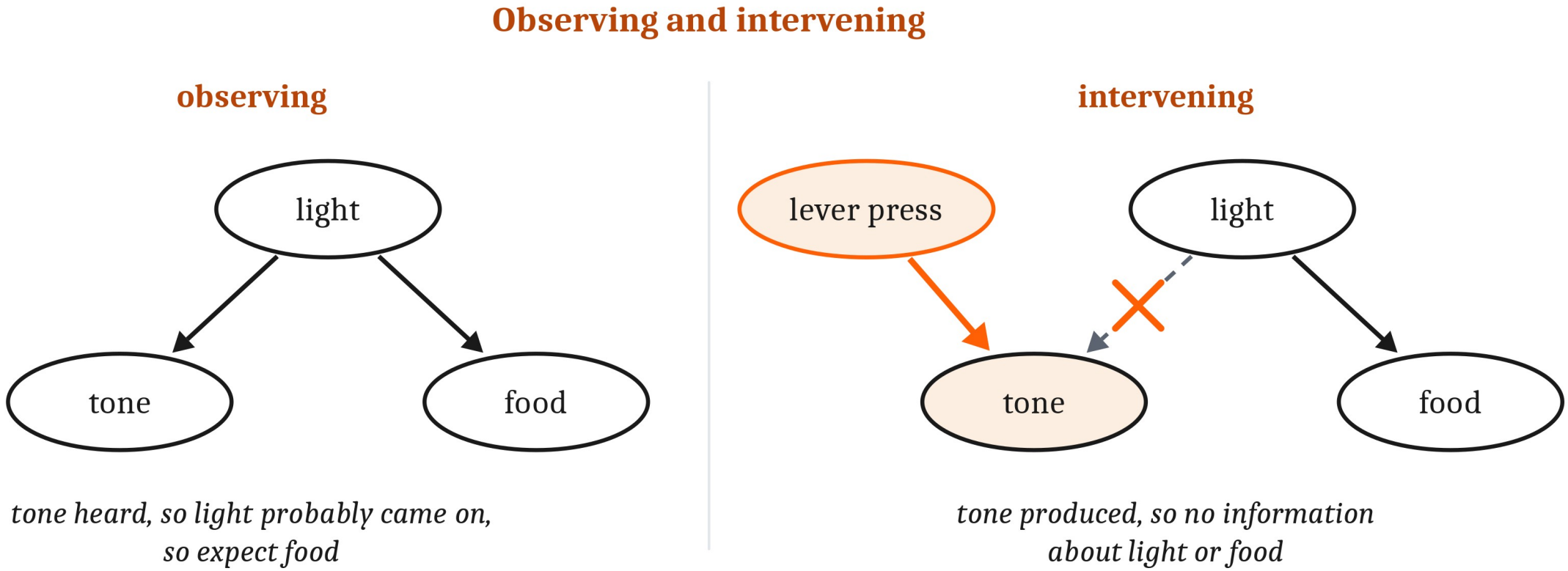
Items like those in Stroop's 1935 experiments

						for 100 items
name the ink color	**RED**	**GREEN**	**BROWN**	**BLUE**	**PURPLE**	**about 110 s**
name the color						**about 63 s**

design and times from Stroop (1935), pp. 647–650; averages of 110.3 and 63.3 seconds, Table III

Reasoning About Causes

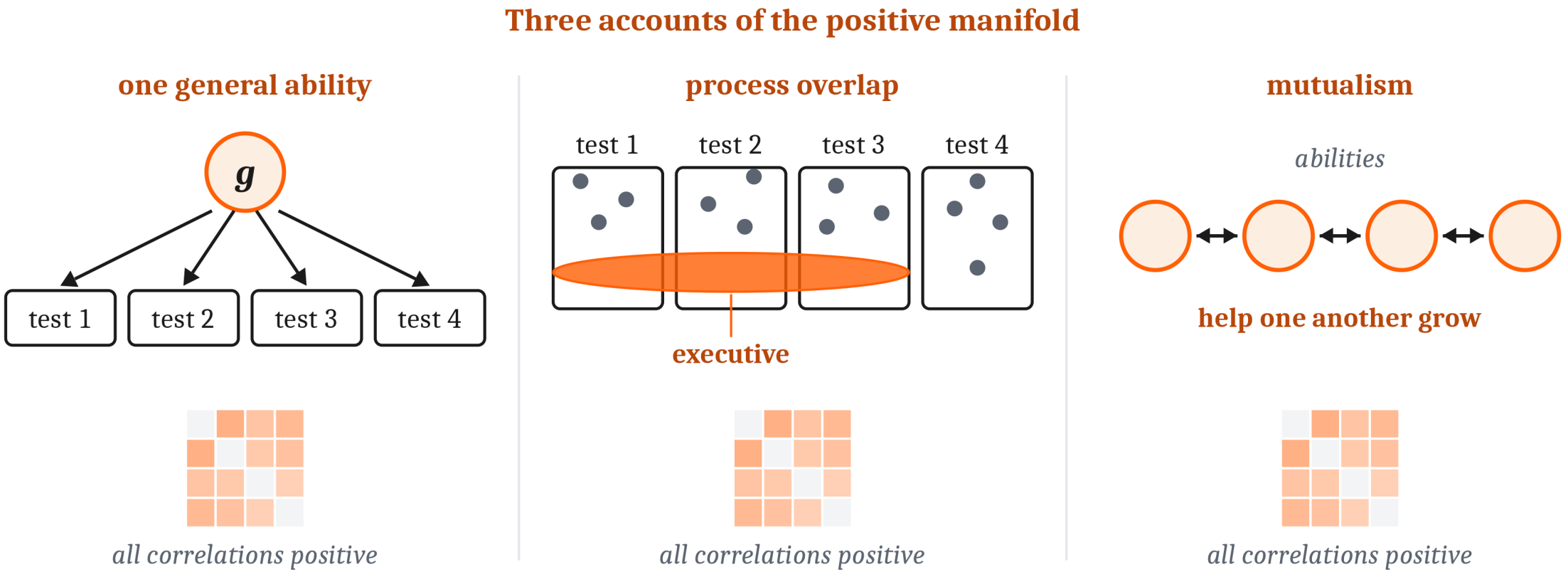
An **intervention** sets the value of one thing from outside, so that whatever usually causes it no longer does. A *causal graph* is a standard for correct causal reasoning, as expected value is a standard for choice.



after Blaisdell et al. (2006), p. 1021, and Gopnik et al. (2004), p. 12

What Produces the Positive Manifold?

General intelligence (g): a proposed single ability that contributes to performance on every mental test. **Process overlap theory**: the proposal that each test draws on several processes, with executive processes shared by most tests. **Mutualism**: the proposal that separate abilities help one another grow during development, so that they end up correlated.



after Kovacs and Conway (2016) and van der Maas et al. (2006); the grids are schematic

General Intelligence in Other Animals, Hominins, and Machines

Cultural intelligence hypothesis: in one version, the proposal that human beings differ from other apes mainly in specialized skills of social cognition, rather than in a uniform advantage across all kinds of problem.

A general factor within one species does not place species on a single scale.

Language models' benchmark scores also correlate, and a factor is not by itself evidence of what produces it.

To Discuss: A Brain-Training App

In your group, five minutes.

A brain-training app advertises that its working-memory games will raise a player’s IQ.

What would have to be true for that claim to hold? Is there evidence that it does?

Image Sources

Slide	Image	File	License	Source
3	Three procedures on the two-step problem	fig-two-step-rungs.png	CC BY 4.0	Jon Willits — from the Module 12 reading · own work
5	Every arrangement of the three-disk Tower of Hanoi	fig-hanoi-states.png	CC BY 4.0	Jon Willits — from the Module 12 reading · own work
8	A value function of the kind proposed in prospect theory	fig-value-function.png	CC BY 4.0	Jon Willits — from the Module 12 reading, after Kahneman and Tversky (1979) · own work
10	Items like those in the original Stroop experiments	fig-stroop.png	CC BY 4.0	Jon Willits — from the Module 12 reading, after Stroop (1935) · own work
11	Observing and intervening in a simple causal graph	fig-observe-intervene.png	CC BY 4.0	Jon Willits — from the Module 12 reading, after Blaisdell et al. (2006) and Gopnik et al. (2004) · own work
12	Three explanations of the positive manifold	fig-manifold-accounts.png	CC BY 4.0	Jon Willits — from the Module 12 reading, after Kovacs and Conway (2016) and van der Maas et al. (2006) · own work